

Jehan Hasan

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RESEARCH INTERESTS

- **Natural Language Processing:** Retrieval-Augmented Generation (RAG) and Agentic AI Applications
- **Green Computing:** Optimization of Long-Term and Short-Term Memory Systems for Energy-Efficient Computing
- **Bioinformatics:** Computational Analysis of Drug–Chemical–Gene Interactions

EDUCATION

Bangladesh University of Engineering and Technology (BUET) , Dhaka, Bangladesh Master of Science in Computer Science and Engineering	Oct 2026 – Present
United International University (UIU) , Dhaka, Bangladesh Bachelor of Science in Computer Science and Engineering Major: Data Science Thesis: Personalized Nutrition Management AI Agent Using LLM Elective Courses: Machine Learning, Digital Image Processing, Bioinformatics, Compiler Design	Feb 2022 – Jun 2026 CGPA: 3.92 /4.00
St. Gregory's High School and College , Dhaka, Bangladesh Higher Secondary Certificate	2017 – 2019
Dhaka Collegiate School , Dhaka, Bangladesh Secondary School Certificate	2009 – 2017

ACADEMIC EXPERIENCE

United International University <i>Undergraduate Teaching Assistant</i>	Dhaka, Bangladesh Jun 2024 – Jun 2026
• Assisted in teaching undergraduate laboratory and theory courses across three consecutive semesters: – Summer 2024: Introduction to Computer Systems – Fall 2024: Data Structure and Algorithms II Laboratory – Fall 2025: Data Structure and Algorithms I Laboratory • Conducted weekly laboratory sessions for approximately 40 undergraduate students, graded assignments and examinations, and provided one-on-one academic support to students. • Developed course materials, prepared lab exercises, and conducted review sessions to enhance student learning.	

DATA SCIENCE PROJECTS

QueryMind <i>Deep Learning Development</i>	[GitHub]
• Developed a production-ready Question Answering system using PyTorch LSTM with deterministic training, Hydra configuration, automated testing, and checkpoint management for reproducible experimentation. • Implemented automated testing, checkpoint management, and a modular pipeline to support reliable model training, evaluation, and deployment.	
Calculus MCP <i>Agentic AI Development</i>	[GitHub]
• Created a production-ready MCP server bringing symbolic calculus capabilities to Claude Desktop. • Implemented differentiation, integration, limit evaluation, and ODE solving using FastMCP, SymPy, and Python.	
MachineTrain Lab <i>Machine Learning, Python, Scikit-learn, Pandas, NumPy, Jupyter</i>	[GitHub]
• Developed a comprehensive Machine Learning repository featuring hands-on implementations of data preprocessing, visualization, feature engineering, and supervised/unsupervised learning algorithms. • Built an end-to-end learning resource with practical Jupyter notebooks covering NumPy, Pandas, Scikit-learn, EDA, model development, and evaluation techniques.	

RESEARCH PROJECTS

AgenticNutrition *(Manuscript Under Review)*

[GitHub]

Agentic AI Development

- Engineered a context-aware AI agent for personalized nutrition management using LangGraph and LangChain.
- Leveraged federated data, EHR-based RAG, and real-time factors to generate disease-specific diet plans via an interactive chatbot.

AdaptiveBeliefMem *(Manuscript Under Review)*

[GitHub]

Memory-Augmented LLM Research

- Proposed a hybrid memory architecture combining implicit belief states with uncertainty-guided explicit memory to improve long-context reasoning while maintaining bounded inference costs.
- Designed an uncertainty-gated memory mechanism that learns when to externalize structured memory, enabling bounded and interpretable reasoning across long conversation trajectories.

FedDDoS

[GitHub]

Machine Learning Based Federated Learning Development

- Developed a federated learning-based DDoS attack detection system for distributed networks.
- Enabled privacy-preserving collaborative threat detection without sharing raw data.

AWARDS

Academic Scholarship - United International University

Dhaka, Bangladesh

- **100% Scholarship:** 4 trimesters (top 2% student)
- **50% Scholarship:** 4 trimesters (top 6% student)
- **25% Scholarship:** 2 trimesters (top 10% student)

Rank 390 - Global Mathematical Talent Probe

Dhaka, Bangladesh

Grade 11

2017

SKILLS

- **Programming Languages:** Python, C, C++ , Java, PHP
- **Data Science Libraries:** LangChain, LangGraph, PyTorch, TensorFlow, FastMCP, Scikit-learn, Pandas, NumPy, Matplotlib
- **Web Frameworks:** Django, Laravel, Flask
- **Database:** SQLite3, MySQL Server
- **Tools & Platforms:** Git, GitHub, Docker, Linux
- **Project Management:** Jira

REFERENCES

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